NAME - KRISH CHANDRASHEKHAR PACHODE ROLL NO. - B-18 BATCH - AIDS (SY-B)

AIM - Implement and analyze various methods of studying correlation and regression including Karl Pearson's coefficient and simple linear regression.

THEORY -

1. What is correlation? How is it different from regression?

Correlation measures the strength and direction of the relationship between two variables. Its value ranges from -1 to +1.

- +1 → Perfect positive correlation
- -1 → Perfect negative correlation
- 0 → No correlation

Regression is used to predict the value of one variable based on another.

Difference:

- Correlation shows relationship (no prediction)
- Regression is used for prediction and modeling

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2. What is simple linear regression? Explain with suitable example.

Simple Linear Regression is a method used to model the relationship between one independent variable (X) and one dependent variable (Y) using a straight line.

Equation: [Y = mX + c]

Where:

- m = slope
- c = intercept

Example: Predicting marks (Y) based on study hours (X).

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3. How do you calculate Pearson correlation in Python?

Pearson correlation can be calculated using Pandas:

```
import pandas as pd

data = {"X": [1,2,3,4], "Y": [2,4,6,8]}
df = pd.DataFrame(data)
```

```
print(df.corr())
```

👉 It gives correlation coefficient between variables.

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4. What is multicollinearity? Does it affect simple regression?

Multicollinearity occurs when independent variables are highly correlated with each other.

👉 It mainly affects **multiple regression**, not simple regression (which has only one independent variable).

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5. A dataset shows strong correlation but poor prediction accuracy. Why?

This can happen because:

- Correlation does not imply causation
- Presence of outliers
- Non-linear relationship
- Overfitting or underfitting
- Missing important variables

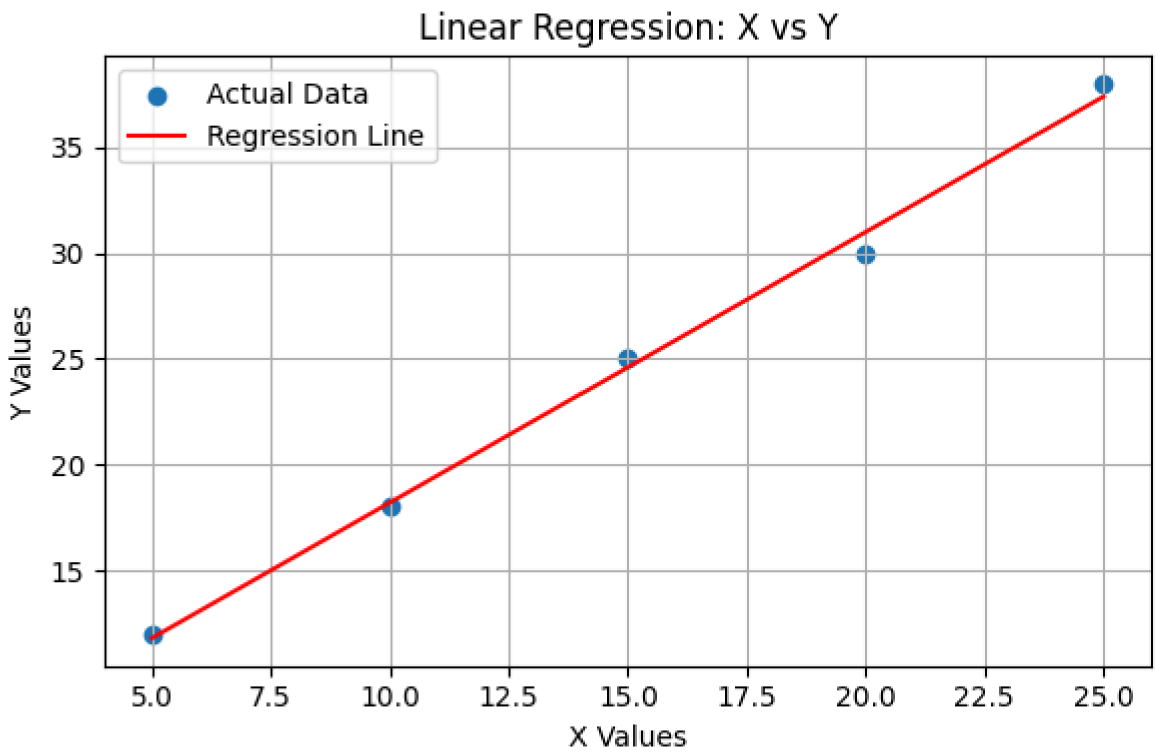
👉 So even with high correlation, prediction may not be accurate.

```
# Import libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import pearsonr
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# -------------------------------
# STEP 1: Sample Data
# -------------------------------
X = [5, 10, 15, 20, 25]
Y = [12, 18, 25, 30, 38]

df = pd.DataFrame({"X": X, "Y": Y})

# -------------------------------
# STEP 2: MANUAL KARL PEARSON
# -------------------------------
def pearson_manual(x, y):
    n = len(x)
    mean_x = np.mean(x)
    mean_y = np.mean(y)
```

```
    numerator = sum((x[i] - mean_x) * (y[i] - mean_y) for i in range(n))
    denominator = (sum((x[i] - mean_x)**2 for i in range(n)) *
                   sum((y[i] - mean_y)**2 for i in range(n))) ** 0.5

    return numerator / denominator

r_manual = pearson_manual(X, Y)
print(f"Karl Pearson Correlation (Manual): {r_manual:.4f}")

# ---------------------------------
# STEP 3: SCIPY VERIFICATION
# ---------------------------------
r_scipy, _ = pearsonr(X, Y)
print(f"Verified using scipy: {r_scipy:.4f}")

# ---------------------------------
# STEP 4: LINEAR REGRESSION
# ---------------------------------
X_df = df[["X"]]   # Independent variable
Y_df = df["Y"]     # Dependent variable

model = LinearRegression()
model.fit(X_df, Y_df)

slope = model.coef_[0]
intercept = model.intercept_

print(f"Regression Equation: Y = {slope:.2f} * X + {intercept:.2f}")

# ---------------------------------
# STEP 5: PREDICTION
# ---------------------------------
y_pred = model.predict(X_df)

# ---------------------------------
# STEP 6: MODEL EVALUATION
# ---------------------------------
mse = mean_squared_error(Y_df, y_pred)
r2 = r2_score(Y_df, y_pred)

print(f"Mean Squared Error (MSE): {mse:.2f}")
print(f"R² Score: {r2:.4f}")

# ---------------------------------
# STEP 7: VISUALIZATION
# ---------------------------------
plt.figure(figsize=(6,4))
plt.scatter(X_df, Y_df, label="Actual Data")
plt.plot(X_df, y_pred, color='red', label="Regression Line")
plt.title("Linear Regression: X vs Y")
plt.xlabel("X Values")
plt.ylabel("Y Values")
plt.legend()
plt.grid(True)
plt.tight_layout()
```

```
plt.show()
```

```
Karl Pearson Correlation (Manual): 0.9981
Verified using scipy: 0.9981
Regression Equation: Y = 1.28 * X + 5.40
Mean Squared Error (MSE): 0.32
R² Score: 0.9961
```



Conclusion – Assignment 8

In this experiment, we studied the concepts of **correlation and regression**. The Karl Pearson correlation coefficient was calculated manually and also verified using the `scipy` library, which showed a strong relationship between the variables.

Further, **simple linear regression** was implemented to model the relationship between the independent and dependent variables. The regression equation was obtained and used for prediction. The performance of the model was evaluated using **Mean Squared Error (MSE)** and **R^2 score**, which indicate the accuracy of the model.

Overall, this experiment helped in understanding how correlation measures the strength of a relationship, while regression is used for prediction and analysis in real-world data.